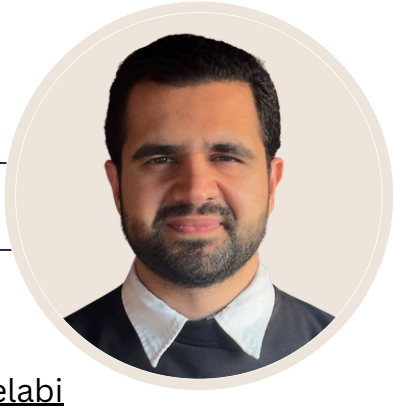
ALI AL-KELABI

AI AND MECHATRONICS ENGINEER



Personal Data

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About me

Master's student in Artificial Intelligence for Smart Sensors and Actuators with hands-on experience in machine learning, computer vision, robotics simulation, data analytics, automation, and software development. Comfortable working across the full development cycle, from data preparation and prototyping to implementation, testing, and documentation. Strong background in Python, TensorFlow/Keras, PySpark, ROS, Docker, Linux, Git, and API-based workflows. Known for a structured, practical, and responsible working style, with a strong interest in using AI to solve real problems efficiently.

Projects

Gesture-Based Access Control System (03.2026 - 04.2026)

- Built a real-time gesture recognition system using webcam input and hand landmarks to classify gestures for access control. Integrated model inference with a GUI and API-based workflow.
- Technologies: Python, MediaPipe, OpenCV, TensorFlow/Keras, GRU, Tkinter, Flask
- Impact: Delivered a usable end-to-end computer vision and ML prototype.

IoT Anomaly Detection – PySpark Pipeline (02.2026 - 03.2026)

- Created a scalable pipeline for processing large sensor datasets and detecting anomalies using statistical methods. Automated preprocessing and output generation in a reproducible workflow.
- Technologies: PySpark, Python, Docker, Linux, Z-score anomaly detection
- Impact: Enabled scalable and containerized analysis of sensor data.

Diabetic Retinopathy Detection (10.2025 - 01.2026)

- Developed an ML/DL-based system to detect diabetic retinopathy and estimate severity from medical image datasets. Managed data preparation, augmentation, model training, and evaluation.
- Technologies: Python, TensorFlow/Keras, image preprocessing, data augmentation
- Impact: Built a practical healthcare AI prototype for image-based classification.

System Communication & Control with Edge Devices (10.2025 - 01.2026)

- Developed a distributed system using two Raspberry Pi devices for communication and control over LoRa. Worked on data flow, local processing, and system integration.
- Technologies: Raspberry Pi, Python, Ubuntu, LoRa, edge computing
- Impact: Gained practical experience with distributed embedded systems and remote communication.

ROS-Based UR5 Robotic Arm Project (10.2025 - 01.2026)

- Implemented a ROS-based UR5 robotic arm system in simulation using ROS1 Kinetic on Ubuntu 16.04. Used a 2D camera with HSV-based color segmentation to detect objects and execute pick-and-place-style motion planning with MoveIt.
- Technologies: ROS1, MoveIt, Gazebo, UR5 simulation, Python/C++, HSV segmentation, 2D camera, Ubuntu 16.04
- Impact: Built a perception-to-action robotics pipeline in simulation, combining computer vision and motion planning.

Local Research Assistant (PyQt6, Qwen2.5 RAG) (02.2026 - 03.2026)

- Developed a PyQt6 desktop chatbot application that analyzes documents (PDF/TXT), generates responses using a local Qwen2.5-0.5B model, and utilizes retrieval-augmented generation.
- Implemented an OpenAI-compatible client, data injection via PyMuPDF, local chat histories, and a two-process architecture (LLM server + GUI), fully offline.
- Demonstrates skills in data integration from files, data flow design, and buildable, data-driven.

Education

M.Eng. Artificial Intelligence for Smart Sensors and Actuators (03.2025 - Today)

Deggendorf Institute of Technology (THD), Cham, Germany

- Focus: ML, DL, Computer vision, System Design, Edge-Device, Advanced Programming, Big-Data
- GPA (German System): 2,4

B.Eng. Mechatronics, (02.2019 - 02.2023)

Internationale Islamic University Malaysia (IIUM), Gombak, Malaysia

Focus: Automation, Mathematics, Robotics, Software Engineering

- GPA (German System): 2,0

Experience

Student Assistant – UiPath & C# Automation of Computer Tasks (06.2025 - 11.2025)

Deggendorf Institute of Technology, Cham, Germany

- Development of a UiPath automation course with practical use cases
- Implementation of web scraping solutions for automated data pipelines
- Focus on process optimization through automation

Working Student – Machine Learning & Sensor Data Analysis (11.2019 - 02.2023)

IIUM & Malaysian Nuclear Agency, Malaysia

- Conducting machine learning research for corrosion detection based on sensor data
- Developing data analysis workflows and methods for anomaly detection
- Bachelor’s thesis: Non-destructive material testing using sensor analysis

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Technical Skills

- Programming: Python, C#, Bash, basic JavaScript
- AI / Machine Learning: TensorFlow, Keras, PyTorch familiarity, computer vision, sequence classification, anomaly detection, data preprocessing, model evaluation
- Data / Analytics: PySpark, Pandas, NumPy, structured data analysis, reporting-style output, automation of data workflows
- Robotics / Engineering: ROS1, ROS2 familiarity, UR5 simulation, MoveIt, Gazebo, motion planning, robot control, HSV-based vision
- Tools / Platforms: Git, Docker, Linux, Ubuntu, Flask, FastAPI familiarity, Jupyter, Raspberry Pi
- Work Style: Problem solving, rapid prototyping, clean documentation, responsible use of AI tools, collaboration in interdisciplinary teams

Languages

- German: C1, English: C2, Arabic: Native

Cham, 20.04.2026